

WILKES COUNTY AP, NC, USA

WMO: 723165

Lat: 36.223N

Lon: 81.098W

Elev: 1300

StdP: 14.02

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63821

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.0	(c) 21.1	(d) -2.5	(e) 5.0	(f) 29.1	(g) 1.1	(h) 6.1	(i) 29.6	(j) 19.9	(k) 39.5	(l) 17.9	(m) 40.2	(n) 3.8	(o) 310	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.4	(c) 91.5	(d) 73.4	(e) 90.2	(f) 72.6	(g) 88.1	(h) 71.8	(i) 77.1	(j) 86.8	(k) 75.7	(l) 84.7	(m) 74.6	(n) 83.1	(o) 4.7	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	136.0	80.9	73.0	128.5	79.0	72.3	125.6	78.4	41.4	86.5	40.2	83.5	38.9	83.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.4	(b) 13.0	(c) 11.1	(e) 10.4	(f) 94.7	(g) 6.1	(h) 3.5	(i) 6.0	(j) 97.2	(k) 2.5	(l) 99.3	(m) -1.0	(n) 101.2	(o) -5.4	(p) 103.8		
(a) 16.4	(b) 13.0	(c) 11.1	(e) 9.1	(f) 78.1	(g) 5.8	(h) 2.0	(i) 4.9	(j) 79.6	(k) 1.6	(l) 80.8	(m) -1.7	(n) 81.9	(o) -5.9	(p) 83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	38.9	41.7	49.1	57.9	65.9	73.3	76.1	74.9	69.1	58.7	48.5	42.0		
	DBStd	15.06	9.68	8.51	9.16	7.87	6.47	4.80	4.08	4.81	5.68	7.63	8.29	7.87		
	HDD50	1180	362	256	131	21	0	0	0	0	0	14	125	271		
	HDD65	3725	809	654	495	233	71	4	0	2	29	221	495	712		
	CDD50	4140	18	23	105	258	493	700	808	772	573	285	81	24		
	CDD65	1207	0	0	3	20	98	254	343	308	152	27	2	0		
	CDH74	10429	2	1	59	312	908	2242	2938	2540	1140	256	29	2		
	CDH80	3802	0	0	5	59	235	872	1212	1025	344	48	2	0		
(14) Wind	WSAvg	3.2	3.8	4.0	4.5	4.4	3.2	2.6	2.1	2.0	2.4	2.8	3.1	3.0		
(15) Precipitation	PrecAvg	49.1	4.3	3.1	4.9	4.0	3.8	4.6	4.6	4.8	4.2	3.6	3.1	4.0		
	PrecMax	63.4	8.6	6.6	8.6	8.3	9.8	8.3	7.0	10.7	10.2	11.5	7.2	8.3		
	PrecMin	36.7	1.1	1.2	1.5	0.8	0.8	1.4	3.0	1.2	0.4	0.1	0.7	1.3		
	PrecStd	7.3	2.0	1.4	1.9	2.3	2.0	2.0	1.2	2.5	2.2	2.4	1.6	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.2	71.9	80.8	85.6	89.9	95.2	94.9	96.9	90.4	84.5	78.9	72.0		
		MCWB	56.4	60.2	61.5	63.4	69.3	73.1	75.6	73.8	69.0	68.8	60.9	59.6		
	2%	DB	63.7	65.9	74.5	81.4	85.9	91.0	91.2	91.2	87.7	80.8	72.3	63.7		
		MCWB	52.9	53.5	58.4	61.4	68.4	72.5	74.2	73.6	69.4	65.3	57.3	55.2		
	5%	DB	59.4	62.5	70.4	78.6	81.9	88.4	90.0	89.7	83.9	76.7	68.1	59.4		
		MCWB	51.3	51.0	55.6	60.8	66.7	71.2	73.6	72.7	69.1	62.6	56.9	53.3		
	10%	DB	54.6	56.8	64.5	73.0	79.4	85.6	87.7	86.1	81.2	72.4	63.7	55.0		
		MCWB	47.1	45.9	52.6	58.5	65.7	70.7	72.9	72.7	68.3	61.8	54.0	48.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	62.7	64.2	68.2	73.2	76.4	79.2	79.2	75.2	73.2	65.1	63.2		
		MCDB	65.0	66.9	75.4	75.6	82.8	88.4	89.1	89.1	82.1	78.7	69.8	64.4		
	2%	WB	56.7	58.4	61.1	65.3	71.1	75.0	77.3	77.2	73.5	70.1	62.5	58.8		
		MCDB	62.5	63.8	69.2	76.1	79.8	86.0	86.9	87.0	80.9	74.1	67.8	62.4		
	5%	WB	52.5	52.8	58.8	63.3	69.4	73.6	76.0	75.5	71.9	67.3	59.5	54.3		
		MCDB	57.1	58.6	66.9	71.6	78.1	83.8	85.1	83.7	79.0	73.5	65.4	58.1		
	10%	WB	48.2	47.9	55.3	61.2	67.9	72.2	74.8	74.2	70.3	64.1	55.4	50.3		
		MCDB	54.1	56.1	62.6	69.4	76.7	81.6	83.3	81.9	76.9	69.0	62.3	54.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.3	19.9	21.4	23.0	22.0	21.9	20.4	20.1	20.2	22.1	22.8	18.4		
		MCDBR	26.1	27.7	28.3	29.8	26.2	26.7	24.4	25.0	26.0	27.6	29.1	24.5		
		MCWBR	17.3	17.6	14.9	13.9	11.9	10.4	9.3	9.2	10.4	13.5	16.4	16.1		
	5% WB	MCDBR	21.4	23.8	23.5	23.7	21.9	23.3	21.0	20.6	20.4	20.5	21.5	20.0		
		MCWBR	16.5	17.8	14.7	13.0	10.8	10.3	9.0	9.0	10.0	11.6	15.3	15.5		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.314	0.343	0.393	0.436	0.471	0.511	0.509	0.403	0.358	0.317	0.302		
	taud		2.499	2.467	2.426	2.293	2.193	2.133	2.047	2.031	2.320	2.434	2.513	2.537		
	Ebn at Noon		284	294	294	282	270	260	248	246	270	274	277	278		
	Edh at Noon		26	30	34	41	46	49	53	53	38	31	25	23		
(44) All-Sky Solar Radiation	RadAvg		779	982	1326	1656	1821	1954	1817	1679	1427	1159	877	669		
	RadStd		54	114	84	131	144	140	126	85	132	141	76	61		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+270	+164

Nomenclature: See separate page